

Total No. of Questions : 8]

SEAT No. :

PB-2253

[Total No. of Pages : 2

[6263]-91

B.E. (Computer Engineering)

MOBILE COMPUTING

(2019 Pattern) (Semester - VII) (410245C) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Define and explain GSM Bust with diagram. [5]
b) Draw and explain the GPRS protocol stack. [5]
c) Explain spectrum allocation with the help of neat diagram. [7]

OR

- Q2)** a) Explain different architectures of WLAN. [5]
b) Explain UTRA-Network (UTRAN) architecture. [5]
c) Explain various traffic and control channels used in GSM network. [7]

- Q3)** a) Explain difference between Hard and Soft Handoff. [5]
b) Explain the Cell dragging in detail. [5]
c) Explain the process of call origination and call termination in GSM. [7]

OR

- Q4)** a) Explain mobility management with neat diagram. [5]
b) Define Handover. List and explain the types of handover. [5]
c) Explain GSM interfaces and GSM protocol architecture. [7]

P.T.O.

- Q5) a)** Describe data transfer from a mobile node to a fixed node and vice versa. [6]
b) Explain agent advertisement and discovery registration in mobile network. [6]
c) Why and how can optimization in mobile IP be achieved? [6]

OR

- Q6) a)** How is packet delivery achieved to and from mobile node. [6]
b) Discuss how tunneling works for mobile IP using IP_in_IP encapsulation. [6]
c) Explain the principal of snooping TCP. State advantages and disadvantages of Snooping TCP. [6]

- Q7) a)** Write short note on VoLGA architecture. [6]
b) Explain evolution from UMTS to LTE. [6]
c) Explain forward W-CDMA channel and reverse W-CDMA channel. [6]

OR

- Q8) a)** Explain in detail Frame format of TD-SCDMA and explain disadvantages of TD-SCDMA. [6]
b) What is HSPA? Explain in detail. [6]
c) Draw and explain functional architecture of 5G. [6]

